

MSDM 5003 Lecture 7-8

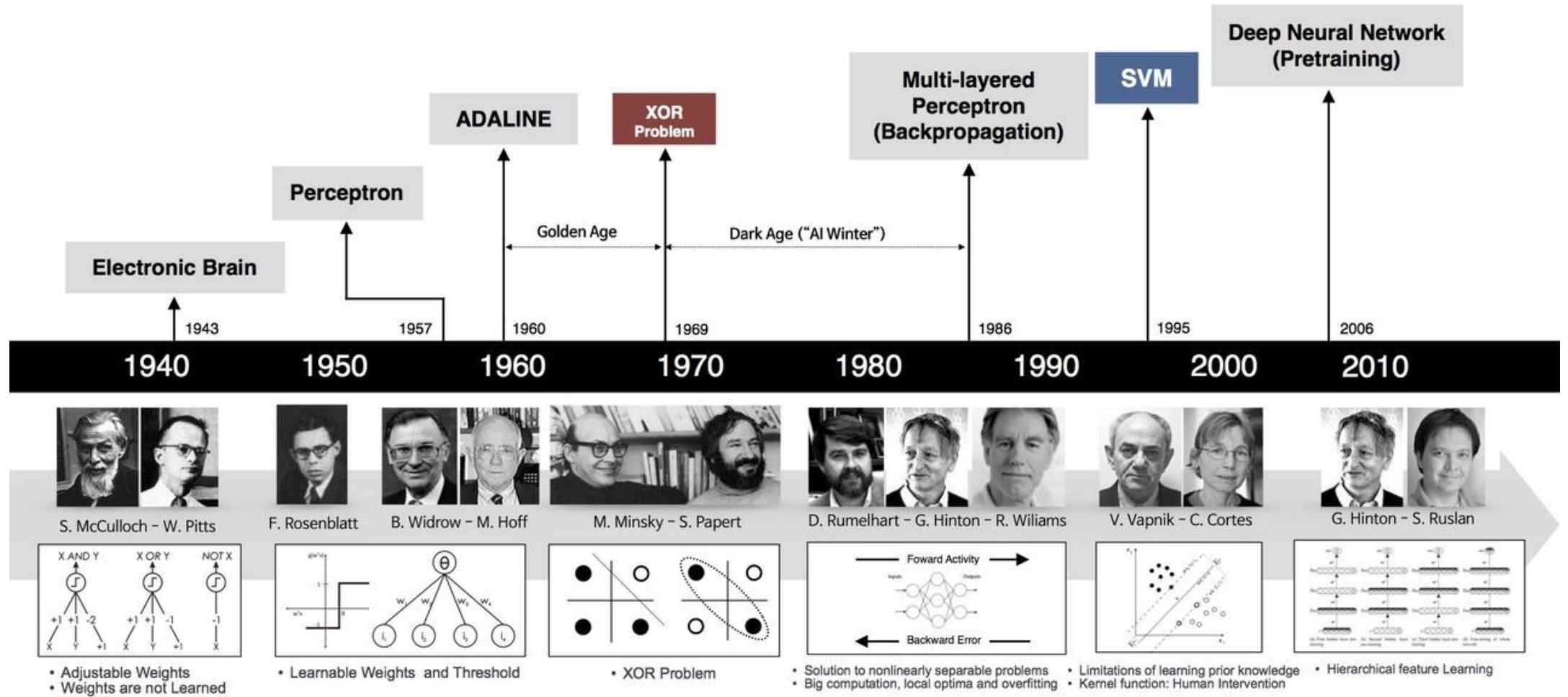
16-23 October 2025

Learning in Neural Networks

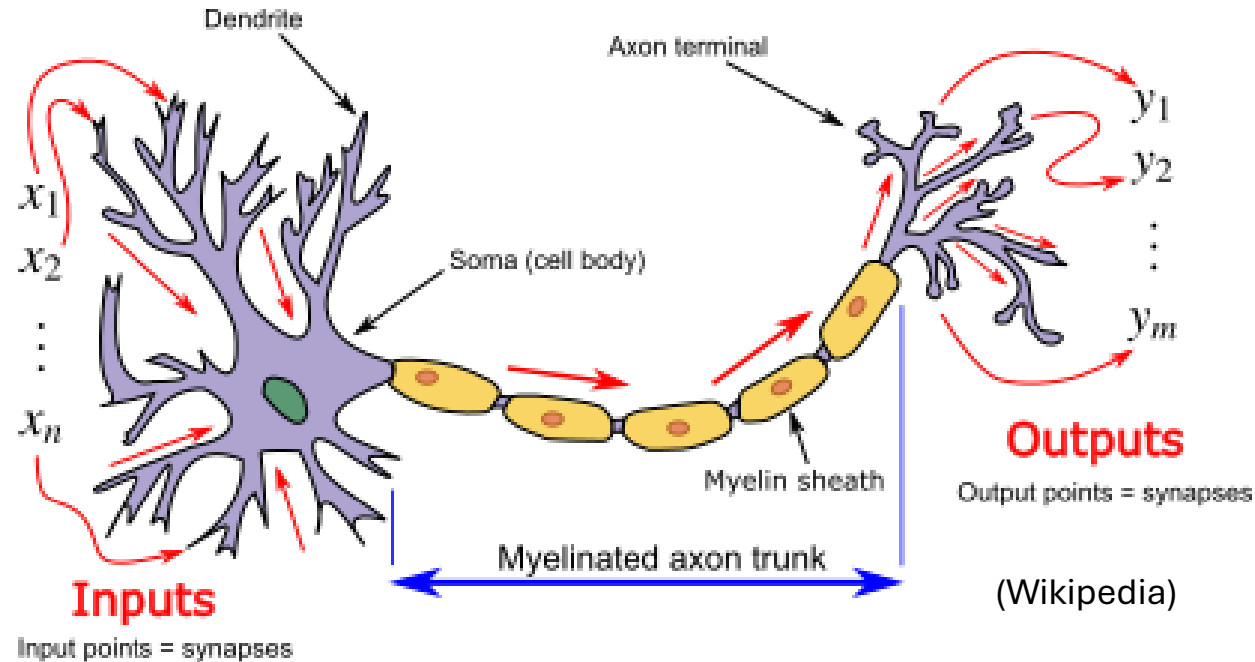
Outline:

1. Brief history of neural networks
2. Learning dynamics in single-layer perceptron
3. Online learning in a perceptron
4. Online learning in a soft committee machine

1. Brief History of Neural Networks

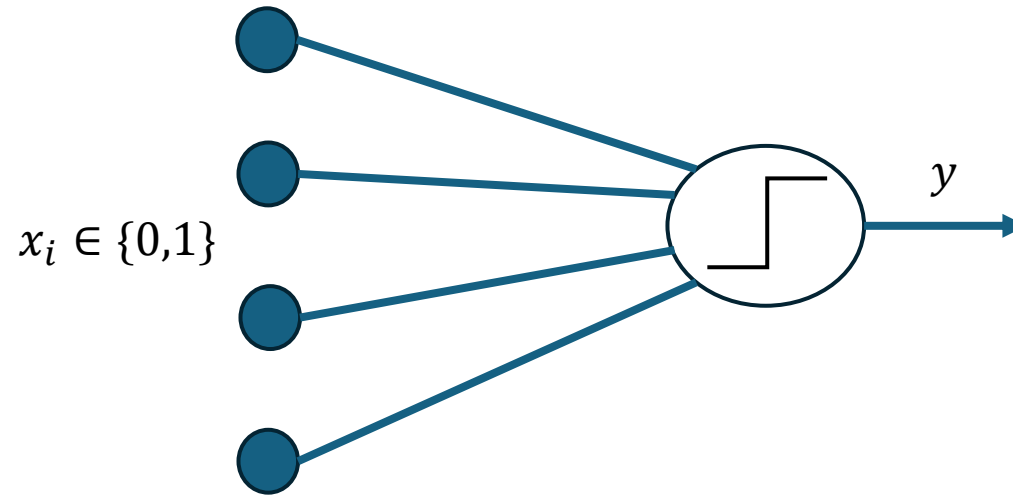


Mystery of the Brain



- How the brain operates has been a **fascinating** topic for scientists.
- Even today, our understanding of the brain is limited, and its operation is certainly **different from traditional computers**.
- The brain consists of 10^{11} **neurons**. Information is carried by **spikes**. Each neuron receives spikes from other neurons through **synapses** and passes information onward to other neurons. The brain consists of 10^{14} synapses.

McCulloch-Pitts Neuron (1943)

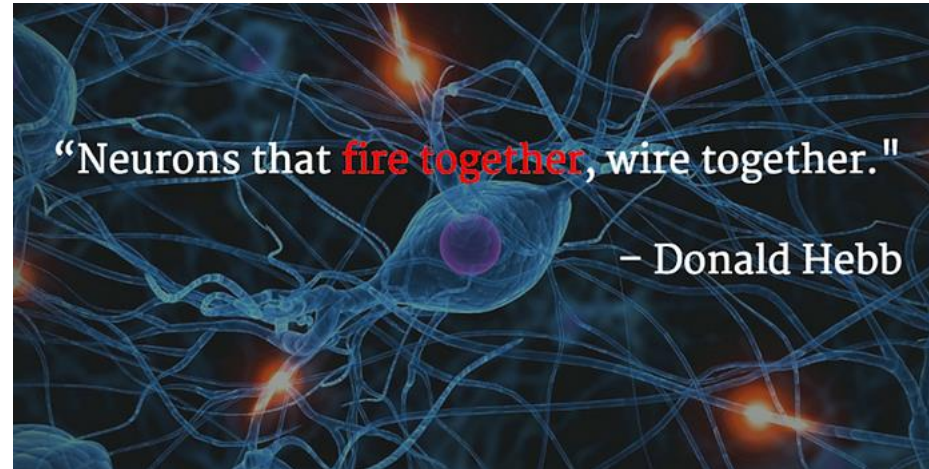


- The state of a neuron can be 1 meaning that it is **firing**, 0 means it is **quiescent**.
- The output is given by

$$y = \Theta \left(\sum_{i=1}^N x_i - \theta \right).$$

- θ is referred to as the **threshold**. Θ is the **step function**, $\Theta(x) = 1$ and 0 otherwise.
- Can be modeled in electrical circuits.

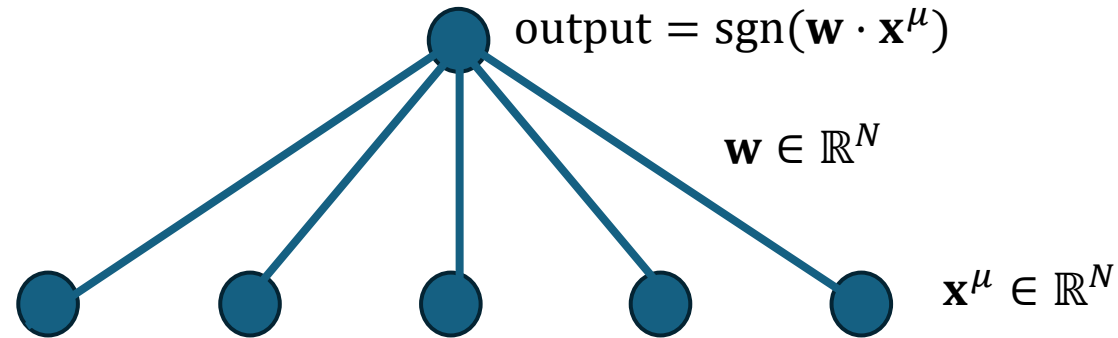
Hebb Rule (1949)



Reut Dayan, "Hebbian Learning:
Biologically Plausible Alternative to
Backpropagation"

- Hebb proposed that if two neurons fire at the same time, the connection between them is strengthened.
- This means that **learning** is possible in the neural network.
- This also means that the correlation between the neuronal states is stored in the **synapses**.
- The property that synapses can change their strengths is called **synaptic plasticity**.

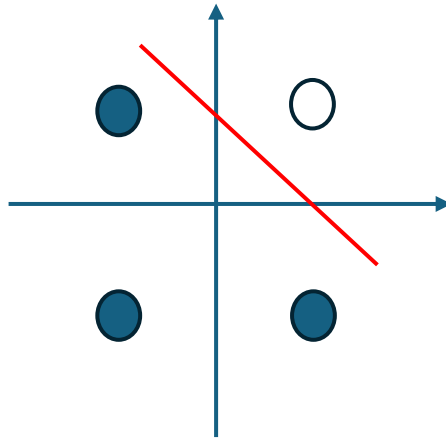
Perceptron (1958)



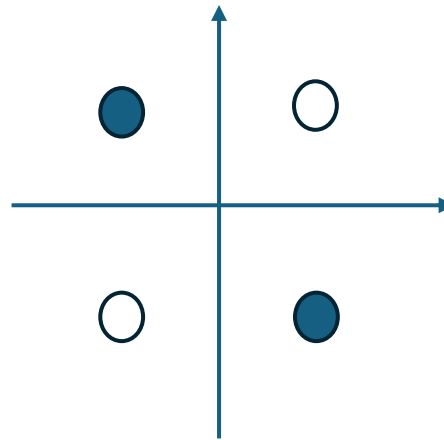
- Rosenblatt proposed the single-layer perceptron with
 - Inputs and weights
 - Activation function (e.g. a step function)
 - Learning algorithm
- **Perceptron learning algorithm:** When an example $(\mathbf{x}^\mu, y^\mu)$ is presented to the perceptron:
 - $\Delta \mathbf{w} = 0$ if the perceptron predicts the correct output
 - $\Delta \mathbf{w} = y^\mu \mathbf{x}^\mu$ if the perceptron predicts the incorrect output
- The perceptron learning algorithm can be proved to be **convergent**.
- The perceptron found applications in fields such as image processing etc.

AI Winter (After 1969)

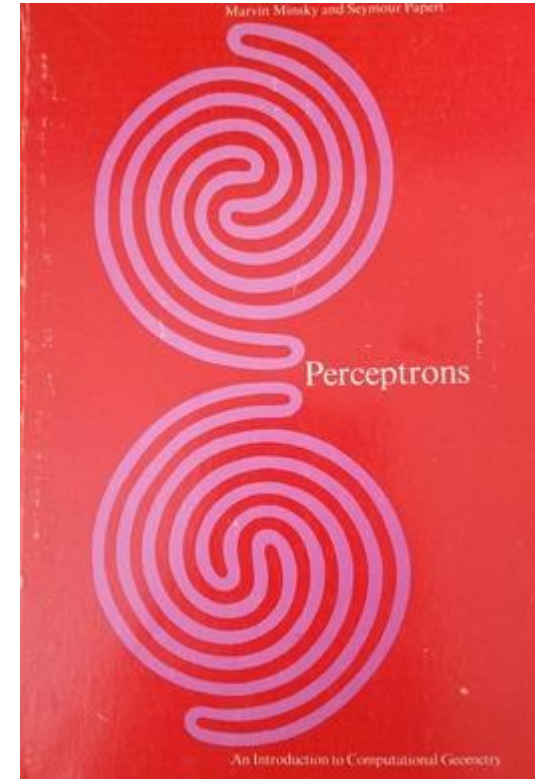
- In 1969, Minsky and Papert published the book *Perceptrons*, discussing the limitations of the perceptron.
- The (single-layer) perceptron can solve **linearly separable** problems, but it cannot solve linearly non-separable problems e.g. the XOR problem.
- In the following decade, neural network research cooled down until it was realized that multi-layer perceptrons had greater processing power.



A separable problem



XOR (non-separable)



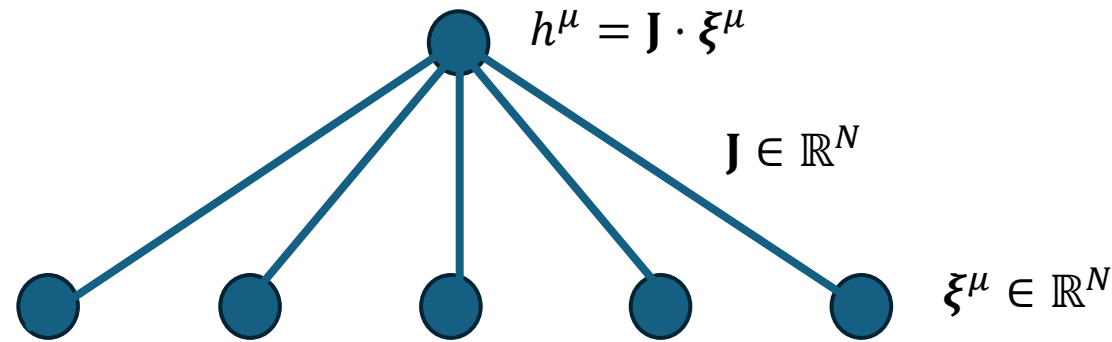
By Marvin Minsky, Seymour Papert -
<https://www.amazon.com/Perceptrons-Introduction-Computational-Marvin-Minsky/dp/0262630222>.

Revival (1980s)

- In the 1980s, there was a revival of interest in neural network research.
- The **backpropagation algorithm** became a popular technique to train multi-layer perceptrons (Rumelhart, Hinton and Williams).
- The Hopfield network became a prototypical model of **associative memory**. It also stimulated the attention of **statistical physicists** on machine learning.
- Both Hopfield and Hinton received the Nobel Prize in Physics 2024.
- The early contribution by Amari on the associative memory should be mentioned.



2. Learning Dynamics in Single-Layer Perceptron



- Consider a single-layer perceptron with N inputs.
- The weight vector of the perceptron is $\mathbf{J} \in \mathbb{R}^N$.
- It is given a set of examples (ξ^μ, y^μ) , $\mu = 1, \dots, p$, and $\xi^\mu \in \mathbb{R}^N$ is the input.
- The learning process is implemented by **minimizing** cost (energy) function

$$E = \sum_{\mu=1}^p f(h^\mu, y^\mu), \quad h^\mu = \mathbf{J} \cdot \xi^\mu.$$

- The minimization is carried out by **gradient descent**,

$$\frac{d\mathbf{J}}{dt} = -\frac{\partial E}{\partial \mathbf{J}}.$$

Gradient Descent and Langevin Equation

- In practice, examples may not be perfect. Noise is often added for improvement.

- **Gradient descent with noise** in the form of **Langevin equation**:

$$\frac{d}{dt}J_j(t) = -\frac{\partial E}{\partial J_j}(t) + \eta_j(t).$$

- $\eta_j(t)$ is the noise at input j and time t , satisfying
$$\langle \eta_j(t) \rangle = 0, \quad \langle \eta_j(t) \eta_k(t') \rangle = 2T \delta_{jk} \delta(t - t'),$$

- T is the **noise temperature**.

- Numerically, this dynamics is carried out by the Euler method:

$$J_j(t + \Delta t) = J_j(t) + \Delta t \left[-\frac{\partial E}{\partial J_j}(t) + \eta_j(t) \right],$$
$$\langle \eta_j(t) \rangle = 0, \quad \langle \eta_j(t) \eta_k(t') \rangle = \frac{2T}{\Delta t} \delta_{jk} \delta_{t,t'}.$$

Methods of Solving Dynamical Equations

- See also Notes2B by Prof. Qian.
- We will use the Ornstein-Uhlenbeck (OU) process as an example.

OU Process for Single Variable

- Stochastic differential equation:

$$dx = -\theta x dt + \sigma dW.$$

- W denotes the Wiener process.
- Often written as a Langevin equation:

$$\frac{dx}{dt} = -\theta x + \sigma \eta(t).$$

- $\eta_j(t)$ is the **white noise** at time t , satisfying
$$\langle \eta(t) \rangle = 0, \quad \langle \eta(t) \eta(t') \rangle = 2T \delta(t - t'),$$

- We have set $\sigma^2 = 2T$; T is the **noise temperature**.

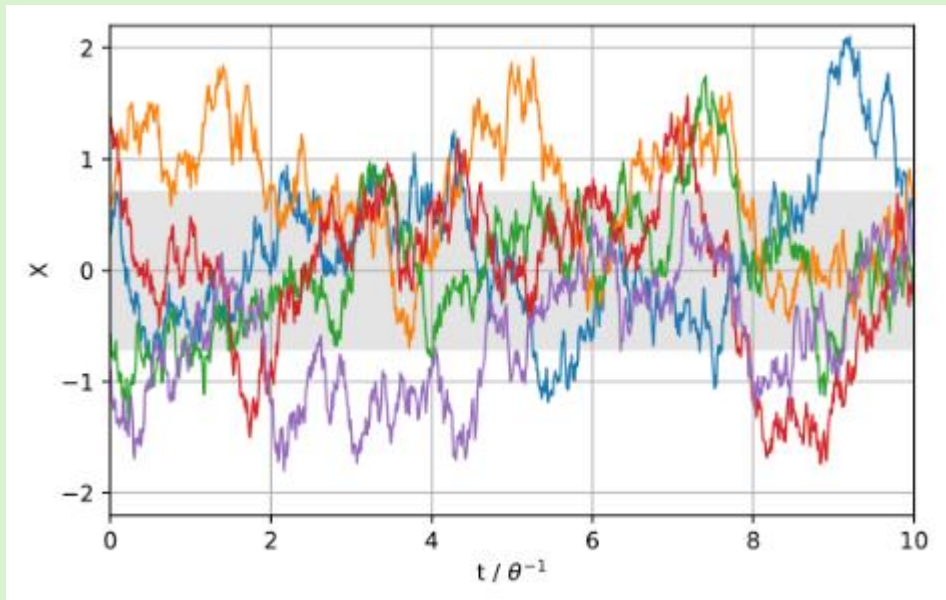
- Numerically, this dynamics is carried out by the Euler method:

$$x(t + \Delta t) = x(t) + \Delta t [-\theta x(t) + \eta(t)],$$
$$\langle \eta(t) \rangle = 0, \quad \langle \eta(t) \eta(t') \rangle = \frac{2T}{\Delta t} \delta_{t,t'}.$$

OU Process

Applications:

- Brownian motion
- Thermal fluctuations
- Stock price volatility



Method 1: Green's Function

First solve the **Green's function** equation:

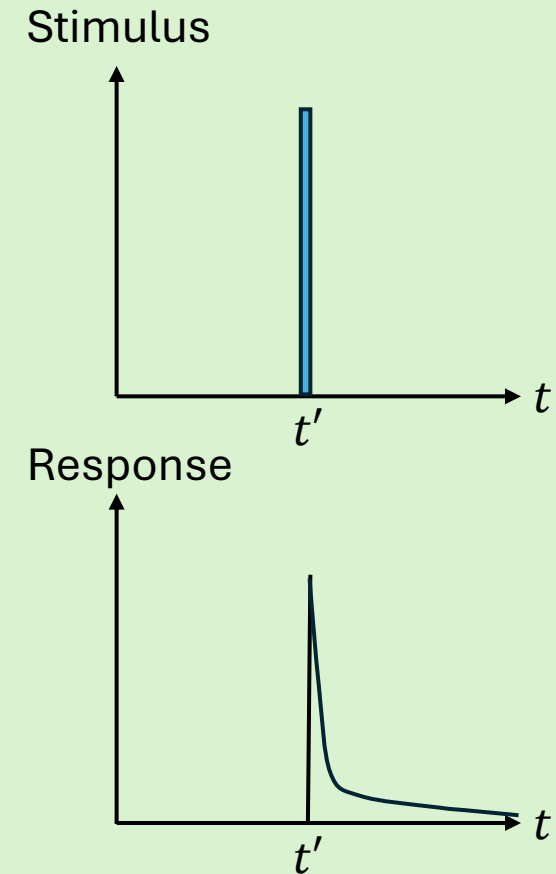
$$\frac{d}{dt} G(t, t') = -\theta G(t, t') + \delta(t - t').$$

Initial condition: $G(t, t') = 0$ for $t < t'$.

The delta function satisfies

$$\int_{-\infty}^{\infty} dt' \delta(t - t') = 1,$$
$$\int_{-\infty}^{\infty} dt' \delta(t - t') f(t') = f(t) \quad \text{for any function } f.$$

The Green's function $G(t, t')$ is the response of the system when it is given a stimulus at time t' .



Solution of the Green's Function Equation

Solution of the **Green's function** equation:

When $t > t'$,

$$\frac{d}{dt} G(t, t') = -\theta G(t, t').$$

Solution: $G(t, t') = Ae^{-\theta t}$.

Integrating both sides of the Green's function equation:

$$\int_{t'-\epsilon}^{t'+\epsilon} dt \frac{d}{dt} G(t, t') = -\theta \int_{t'-\epsilon}^{t'+\epsilon} dt G(t, t') + \int_{t'-\epsilon}^{t'+\epsilon} dt \delta(t - t').$$

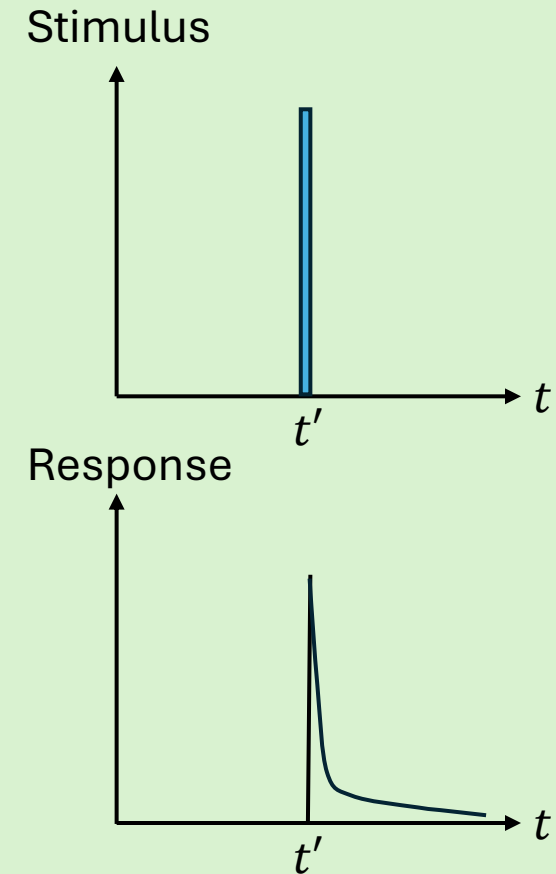
$$\text{LHS} = G(t' + \epsilon, t') - G(t' - \epsilon, t') = Ae^{-\theta t'} - 0 = Ae^{-\theta t'}.$$

$$\text{RHS} = 0 + 1.$$

$$\Rightarrow Ae^{-\theta t'} = 1 \Rightarrow A = e^{\theta t'}.$$

Solution:

$$G(t, t') = \Theta(t - t')e^{-\theta(t-t')}.$$



Integrating the Responses

Multiplying the **Green's function** equation by $\eta(t')$ and integrating,

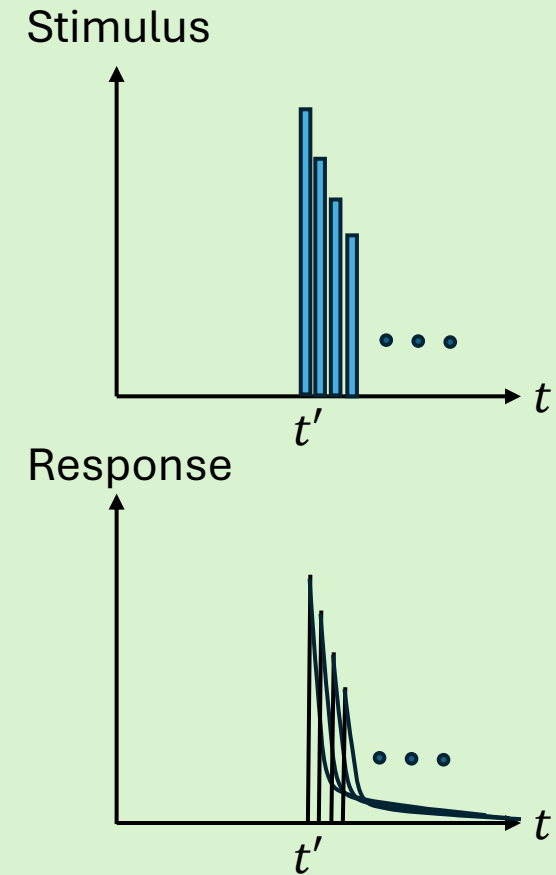
$$\begin{aligned} & \int_{-\infty}^{\infty} dt' \eta(t') \frac{d}{dt} G(t, t') \\ &= -\theta \int_{-\infty}^{\infty} dt' \eta(t') G(t, t') + \int_{-\infty}^{\infty} dt' \eta(t') \delta(t - t'). \\ \frac{d}{dt} \int_{-\infty}^{\infty} dt' G(t, t') \eta(t') &= -\theta \int_{-\infty}^{\infty} dt' G(t, t') \eta(t') + \eta(t). \end{aligned}$$

This shows that the solution to the OU process is

$$x(t) = \int_{-\infty}^{\infty} dt' G(t, t') \eta(t') = \int_{-\infty}^t dt' e^{-\theta(t-t')} \eta(t').$$

This solution satisfies

$$\langle x(t) \rangle = \int_{-\infty}^t dt' e^{-\theta(t-t')} \langle \eta(t') \rangle = 0.$$



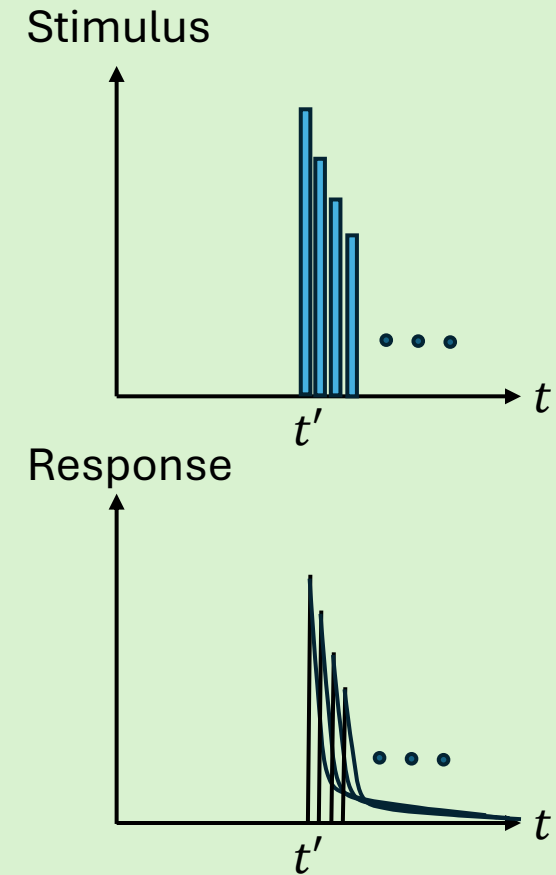
Variance and Distribution

Calculating the variance,

$$\begin{aligned}\langle x(t)^2 \rangle &= \int_{-\infty}^t dt_1 e^{-\theta(t-t_1)} \int_{-\infty}^t dt_2 e^{-\theta(t-t_2)} \langle \eta(t_1) \eta(t_2) \rangle \\ &= 2T \int_{-\infty}^t dt_1 e^{-\theta(t-t_1)} \int_{-\infty}^t dt_2 e^{-\theta(t-t_2)} \delta(t_1 - t_2) \\ &= 2T \int_{-\infty}^t dt_1 e^{-2\theta(t-t_1)} = \frac{T}{\theta}.\end{aligned}$$

Higher order moments can be found (for example, $\langle x(t)^4 \rangle$ in the next slide). They agree with the moments of the Gaussian distribution with mean 0 and variance T/θ . Therefore,

$$P(x) = \frac{\exp\left(-\frac{\theta x^2}{2T}\right)}{\sqrt{2\pi T/\theta}}.$$



Higher Order Moments

$$\langle x(t)^4 \rangle = \int_{-\infty}^t dt_1 e^{-\theta(t-t_1)} \int_{-\infty}^t dt_2 e^{-\theta(t-t_2)} \int_{-\infty}^t dt_3 e^{-\theta(t-t_3)} \int_{-\infty}^t dt_4 e^{-\theta(t-t_4)} \\ \times \langle \eta(t_1) \eta(t_2) \eta(t_3) \eta(t_4) \rangle.$$

Dominant contributions come from pairing $\eta(t_1), \eta(t_2), \eta(t_3), \eta(t_4)$ in pairs. There are 3 combinations: $\langle \eta(t_1) \eta(t_2) \rangle \langle \eta(t_3) \eta(t_4) \rangle$, $\langle \eta(t_1) \eta(t_3) \rangle \langle \eta(t_2) \eta(t_4) \rangle$, and $\langle \eta(t_1) \eta(t_4) \rangle \langle \eta(t_2) \eta(t_3) \rangle$. Each combination has the same contribution. Hence,

$$\langle x(t)^4 \rangle = 3 \left[\int_{-\infty}^t dt_1 e^{-\theta(t-t_1)} \int_{-\infty}^t dt_2 e^{-\theta(t-t_2)} \langle \eta(t_1) \eta(t_2) \rangle \right]^2 = 3 \left(\frac{T}{\theta} \right)^2.$$

In general, we can show that

$$\langle x(t)^{2n} \rangle = (2n-1)!! \left(\frac{T}{\theta} \right)^n \quad \text{and} \quad \langle x(t)^{2n+1} \rangle = 0.$$

These are exactly the higher order moments of the Gaussian distribution.

Method 2: Fokker-Planck Equation

This is done by relating the probabilities $P(x, t + \Delta t)$ and $P(x, t)$ directly. Consider the Langevin equation in discrete time steps,

$$x' = x + \Delta t(-\theta x + \eta).$$

We find the delta function helpful:

$$P(x', t + \Delta t) = \int dx P(x, t) \delta[x + \Delta t(-\theta x + \eta) - x'].$$

We can check that the normalization of the distribution is preserved:

$$\int dx' P(x', t + \Delta t) = \int dx P(x, t) \int dx' \delta[x + \Delta t(-\theta x + \eta) - x'] = \int dx P(x, t) = 1.$$

This is still difficult to handle. Hence, we consider how an arbitrary average of a function $F(x)$ changes with time. Let

$$\bar{F}(t) = \int dx P(x, t) F(x).$$

The Time-Dependent Average

$\bar{F}(t)$ changes with time according to

$$\bar{F}(t + \Delta t) = \int dx' P(x', t + \Delta t) F(x').$$

Substituting the expression of $P(x', t + \Delta t)$,

$$\bar{F}(t + \Delta t) = \int dx' \int dx P(x, t) \delta[x + \Delta t(-\theta x + \eta(t)) - x'] F(x').$$

Using the property of the delta function,

$$\bar{F}(t + \Delta t) = \int dx P(x, t) F[x + \Delta t(-\theta x + \eta(t))].$$

Taylor expansion to second order of Δt ,

$$\bar{F}(t + \Delta t) = \int dx P(x, t) \left[F(x) + \Delta t \frac{\partial F}{\partial x} (-\theta x + \eta(t)) + \frac{\Delta t^2}{2} \eta(t)^2 \frac{\partial^2 F}{\partial x^2} \right].$$

The Fokker-Planck Equation

Using the averages $\langle \eta(t) \rangle = 0$ and $\langle \eta(t)^2 \rangle = 2T/\Delta$,

$$\bar{F}(t + \Delta t) = \int dx P(x, t) \left[F(x) + \Delta t \frac{\partial F}{\partial x} (-\theta x) + (\Delta t) T \frac{\partial^2 F}{\partial x^2} \right].$$

Integrating by parts,

$$\bar{F}(t + \Delta t) = \int dx F(x) \left[P(x, t) + \Delta t \frac{\partial}{\partial x} (\theta x P(x, t)) + (\Delta t) T \frac{\partial^2}{\partial x^2} P(x, t) \right].$$

At the same time, the left hand side becomes

$$\bar{F}(t + \Delta t) = \int dx F(x) \left(P(x, t) + \Delta t \frac{\partial P}{\partial t} \right).$$

Since $F(x)$ is arbitrary, we arrive at the Fokker-Planck equation,

$$\frac{\partial P}{\partial t} = \frac{\partial}{\partial x} (\theta x P(x, t)) + T \frac{\partial^2}{\partial x^2} P(x, t).$$

The Steady State Solution

When the OU process approaches the steady state, $\partial P / \partial t = 0$. Then

$$\frac{\partial}{\partial x} (\theta x P(x)) + T \frac{\partial^2}{\partial x^2} P(x) = 0.$$

Integrating,

$$\theta x P(x) + T \frac{\partial P}{\partial x} = 0.$$

The solution is the Gaussian distribution

$$P(x) \sim \exp\left(-\frac{\theta x^2}{2T}\right)$$

Application to Learning Dynamics

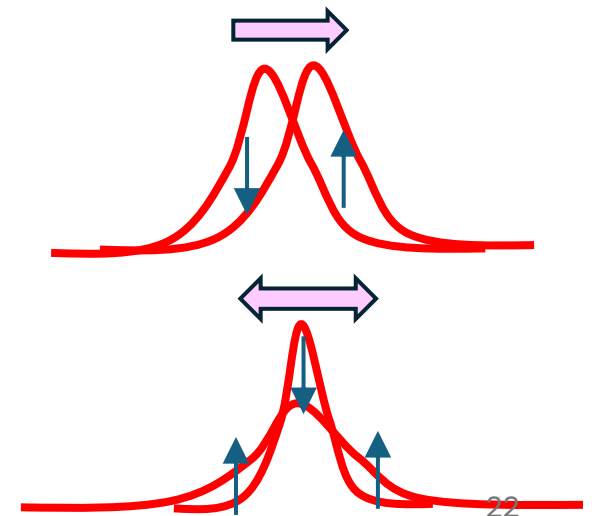
- Applying the **Fokker-Planck equation** to learning dynamics,

$$\frac{\partial P(J, t)}{\partial t} = \frac{\partial}{\partial J} \left(P(J, t) \frac{\partial E(J)}{\partial J} \right) + T \frac{\partial^2 P(J, t)}{\partial J^2}.$$

- This equation tells us that the evolution of the weight distribution is due to two factors:

$$\frac{\partial}{\partial J} \left(P(J, t) \frac{\partial E(J)}{\partial J} \right) \text{ is the drift,}$$

$$T \frac{\partial^2 P(J, t)}{\partial J^2} \text{ is the diffusion.}$$



Boltzmann Distribution

- At the steady state, the time derivative becomes 0. The Fokker-Planck equation becomes

$$\frac{\partial}{\partial J} \left(P(J) \frac{\partial E(J)}{\partial J} \right) + T \frac{\partial^2 P(J)}{\partial J^2} = 0.$$

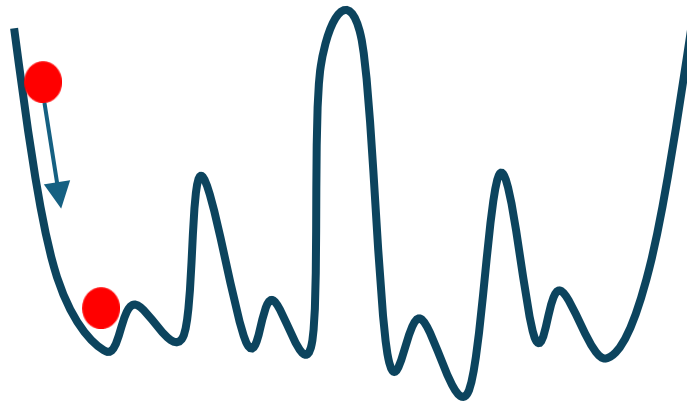
- The solution of this equation is:

$$P(J) = \frac{1}{Z} \exp \left(-\frac{E(J)}{T} \right).$$

- Z is the normalization factor to ensure that $\int dJ P(J) = 1$.
- This shows that the steady state of the gradient descent dynamics is the **Boltzmann distribution at the noise temperature T** .

Complex Energy Landscapes

- In many complex learning problems, the energy may consist of many **local minima**.
- Starting from a typical initial state, the learning process may end up in a local minimum separated from other minima by **energy barriers**.
- **Adding noise** enables the system to overcome the barriers and escape from the local minima.



Effects of Temperature

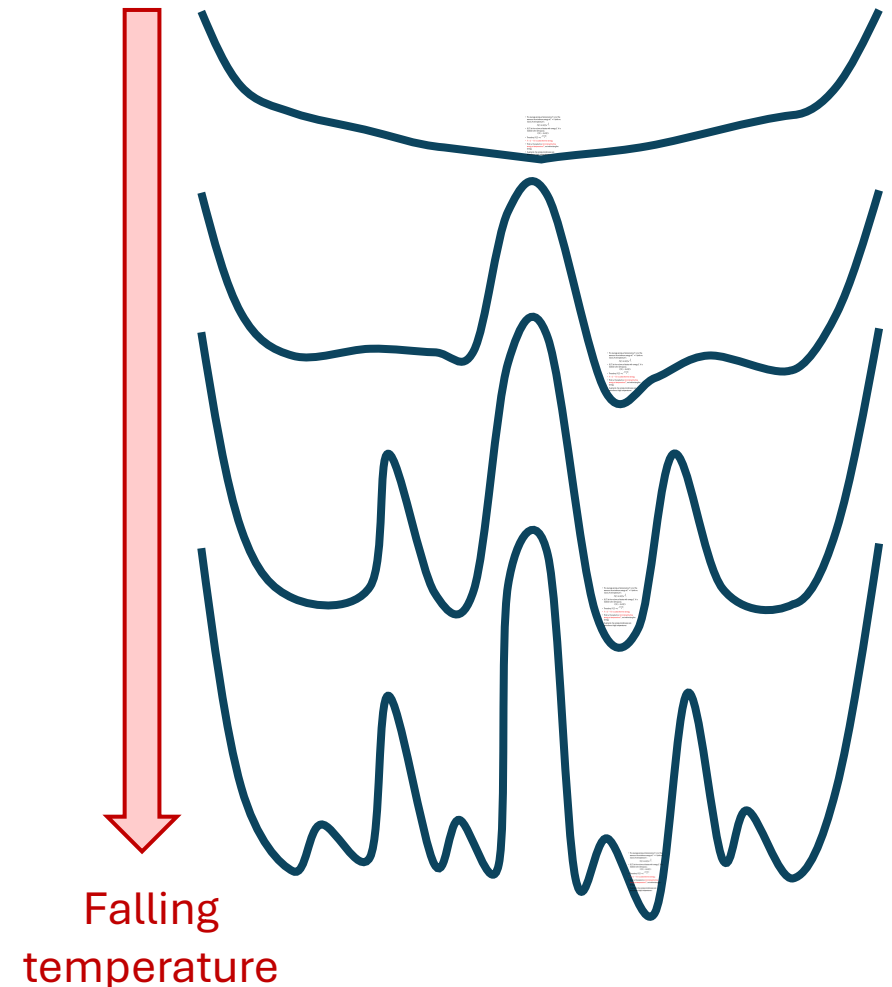
- The average energy at temperature T is not the same as the minimum energy at $T = 0$ (with no noise). At temperature T ,

$$P(E) \propto \Omega(E)e^{-\frac{E}{T}}.$$

- $\Omega(E)$ is the volume of states with energy E . It is related to the entropy by

$$S(E) = \ln \Omega(E) \Rightarrow \Omega(E) = e^{\frac{S(E)}{T}}$$

- Therefore, $P(E) \propto e^{-\left(\frac{E-TS(E)}{T}\right)}$.
- $F = E - TS$ is called the free energy.
- That is, the system is **minimizing the free energy at temperature T** , not minimizing the energy.
- In general, free energy landscapes are smoother at high temperatures.



Stochastic Gradient Descent

- The cost function used for learning is the sum of terms contributed from the entire example set.
- For large datasets, the computation of the gradient at each learning step is **computationally expensive**.
- To reduce the workload, **stochastic gradient descent** (SGD) randomly selects a subset of examples to be included in calculating the gradient term.
- In deep learning, it is a surprise that SGD can even **outperform** full gradient descent.
- The most extreme practice is **online learning**, in which only one randomly selected example is presented in each step.
- Since the examples at different steps are **uncorrelated**, the analysis of online learning is much simpler.

3. On-Line Learning with a Perceptron.

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PACS. 87.10 – General, theoretical, and mathematical biophysics (inc. logic of biosystems, quantum biology and relevant aspects of thermodynamics, information theory, cybernetics, and bionics).

PACS. 02.50 – Probability theory, stochastic processes, and statistics.

PACS. 05.90 – Other topics in statistical physics and thermodynamics.

Abstract. – We study on-line learning of a linearly separable rule with a simple perceptron. Training utilizes a sequence of uncorrelated, randomly drawn N -dimensional input examples. In the thermodynamic limit the generalization error after training such examples with P can be calculated exactly. For the standard perceptron algorithm it decreases like $(N/P)^{1/3}$ for large P/N , in contrast to the faster $(N/P)^{1/2}$ -behaviour of the so-called Hebbian learning. Furthermore, we show that a specific parameter-free on-line scheme, the AdaTron algorithm, gives an asymptotic (N/P) -decay of the generalization error. This coincides (up to a constant factor) with the bound for any training process based on random examples, including off-line learning. Simulations confirm our results.

The Teacher Perceptron

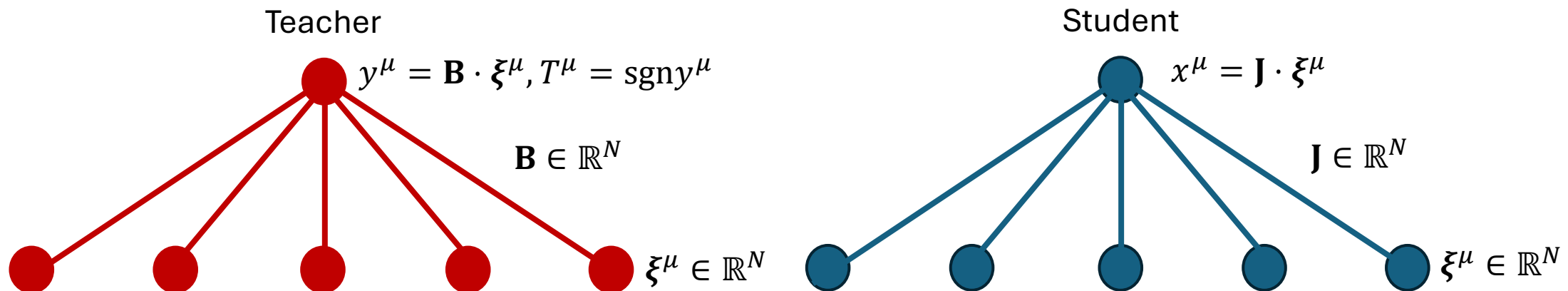
- Consider a linearly separable target rule represented by the **teacher vector** $B_j \in R$ for $j = 1, \dots, N$.
- The teacher vector is normalized $\sum_{j=1}^N B_j^2 = 1$.
- Hence, B_j is typically $O\left(\frac{1}{\sqrt{N}}\right)$.
- At time step μ , a new, uncorrelated input ξ_j^μ ($j = 1, \dots, N$ and $\mu = 1, \dots, p$) is presented, where ξ_j^μ are taken to be i.i.d. random numbers with zero mean and unit variance.
- The teacher perceptron produces the output $T_\mu = \text{sgn} y_\mu$, where $y_\mu = \mathbf{B} \cdot \boldsymbol{\xi}^\mu$.
- Using the central limit theorem, y_μ has a **Gaussian distribution** with mean 0 and variance $\langle y_\mu^2 \rangle = \left\langle \sum_{j,k} B_j \xi_j^\mu B_k \xi_k^\mu \right\rangle = \sum_j B_j^2 \left\langle \left(\xi_j^\mu \right)^2 \right\rangle = \sum_j B_j^2 = 1$.

The Student Perceptron

- The student perceptron is represented by the student vector $J_j \in \mathbb{R}$ for $j = 1, \dots, N$.
- The student output for example μ is given by $S_\mu = \text{sgn} x_\mu$ where $x_\mu = \mathbf{J} \cdot \boldsymbol{\xi}^\mu$.
- The student learns by the rule

$$J_j^{\mu+1} = J_j^\mu + \frac{1}{N} f(T_\mu, x_\mu) T_\mu \xi_j^\mu.$$

- The function f depends on the learning algorithm. For example, $f(T_\mu, x_\mu) = 1$ for Hebb rule, $f(T_\mu, x_\mu) = \Theta(-T_\mu x_\mu)$ for the perceptron learning rule.

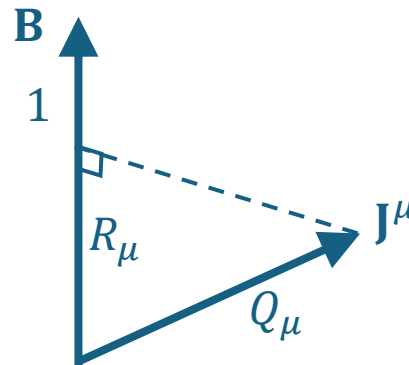


Order Parameters

- To analyze the performance of the learning process, we introduce the following two **order parameters** for large N ,

$$Q_\mu = \sqrt{\mathbf{J}^\mu \cdot \mathbf{J}^\mu},$$
$$R_\mu = \mathbf{J}^\mu \cdot \mathbf{B}.$$

- It is instructive to visualize the geometry of the teacher and student vectors. The target of learning is to make the student vector $\mathbf{J}^\mu$ **align** with the teacher vector $\mathbf{B}$.



Dynamical Equation of Q^μ

- In a single step,

$$J_j^{\mu+1} = J_j^\mu + \frac{1}{N} f_\mu T_\mu \xi_j^\mu.$$

- Squaring both sides and summing over j ,

$$\sum_j \left(J_j^{\mu+1}\right)^2 = \sum_j \left(J_j^\mu\right)^2 + \frac{2}{N} f_\mu T_\mu \sum_j J_j^\mu \xi_j^\mu + \frac{1}{N^2} f_\mu^2 T_\mu^2 \sum_j \left(\xi_j^\mu\right)^2.$$

- Simplifying the expression,

$$\left(Q_{\mu+1}\right)^2 - \left(Q_\mu\right)^2 = \frac{2}{N} f_\mu T_\mu x_\mu + \frac{1}{N} f_\mu^2 T_\mu^2.$$

- On the left-hand side, $\left(Q_{\mu+1}\right)^2 - \left(Q_\mu\right)^2 \approx 2Q_\mu \Delta Q_\mu$. Hence, we have

$$\Delta Q_\mu = \frac{1}{NQ_\mu} f_\mu T_\mu x_\mu + \frac{1}{2NQ_\mu} f_\mu^2.$$

Dynamical Equation of R^μ

- Change of R_μ in a single step,

$$\Delta R_\mu = \sum_j B_j \Delta J_j^\mu = \frac{1}{N} \sum_j B_j f_\mu T_\mu \xi_j^\mu = \frac{1}{N} f_\mu T_\mu y_\mu.$$

- In the limit $N \rightarrow \infty$, we introduce the continuous time $\alpha = \frac{\mu}{N}$. Hence, the changes ΔQ_μ and ΔR_μ take place in a time increment $\Delta\alpha = \frac{1}{N}$. This shows that

$$\frac{\Delta Q_\mu}{\Delta\alpha} = \frac{1}{Q_\mu} f_\mu T_\mu x_\mu + \frac{1}{2Q_\mu} f_\mu^2, \quad \frac{\Delta R_\mu}{\Delta\alpha} = f_\mu T_\mu y_\mu.$$

- Assume that the distributions of x_μ and y_μ evolve slowly in a time scale of $O(1)$. We introduce the joint distribution $P(x, y)$ and arrive at

$$\frac{dQ}{d\alpha} = \int dx dy P(x, y) \left[\frac{1}{Q} f(T, x) T x + \frac{1}{2Q} f(T, x)^2 \right],$$

$$\frac{dR}{d\alpha} = \int dx dy P(x, y) f(T, x) T y.$$

Teacher and Student Fields Are Gaussian

- Since the components of the example inputs are i.i.d., the teacher field is Gaussian distributed with mean 0 and variance

$$\langle y_\mu^2 \rangle = \left\langle \sum_{j,k} B_j \xi_j^\mu B_k \xi_k^\mu \right\rangle = \sum_j B_j^2 \langle (\xi_j^\mu)^2 \rangle = \sum_j B_j^2 = 1.$$

- Similarly, the student field is Gaussian distributed with mean 0 and variance

$$\langle x_\mu^2 \rangle = \left\langle \sum_{j,k} J_j^\mu \xi_j^\mu J_k^\mu \xi_k^\mu \right\rangle = \sum_j (J_j^\mu)^2 \langle (\xi_j^\mu)^2 \rangle = \sum_j (J_j^\mu)^2 = Q_\mu^2.$$

- The covariance between the student and teacher fields is given by

$$\langle x_\mu y_\mu \rangle = \left\langle \sum_{j,k} J_j^\mu \xi_j^\mu B_k \xi_k^\mu \right\rangle = \sum_j J_j^\mu B_j \langle (\xi_j^\mu)^2 \rangle = \sum_j J_j^\mu B_j = R_\mu.$$

Teacher and Student Field Distribution

- We may introduce two independent Gaussian variables u and v to characterize the student and teacher fields,

$$x_\mu = R_\mu u + \sqrt{Q_\mu^2 - R_\mu^2} v,$$
$$y_\mu = u.$$

- These expressions give the correct covariance and variances. For example,

$$\langle x_\mu^2 \rangle = R_\mu^2 \langle u^2 \rangle + 2R_\mu \sqrt{Q_\mu^2 - R_\mu^2} \langle uv \rangle + (Q_\mu^2 - R_\mu^2) \langle v^2 \rangle = Q_\mu^2.$$

- The joint distribution $P(x_\mu, y_\mu)$ is given by

$$P(x_\mu, y_\mu) = \int Du \int Dv \delta(u - y_\mu) \delta\left(R_\mu u + \sqrt{Q_\mu^2 - R_\mu^2} v - x_\mu\right),$$

where $Du \equiv \exp\left(-\frac{u^2}{2}\right) / \sqrt{2\pi}$ is the gaussian measure.

Order Parameter Equations

- Self-contained differential equations in terms of Q and R ,

$$\frac{dQ}{d\alpha} = \int Du \int Dv \left[\frac{1}{Q} f(T, x) \text{sgnu} \left(Ru + \sqrt{Q^2 - R^2} v \right) + \frac{1}{2Q} f(T, x)^2 \right],$$

$$\frac{dR}{d\alpha} = \int Du \int Dv f(T, x) (\text{sgnu})(u),$$

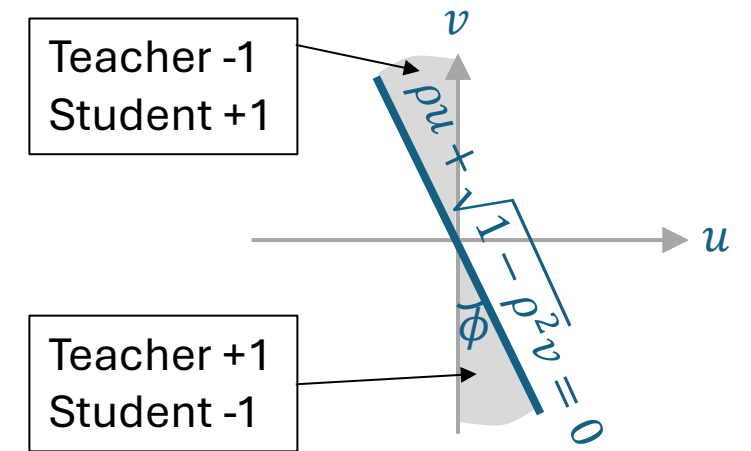
where $f(T, x) = f \left[\text{sgnu}, Ru + \sqrt{Q^2 - R^2} v \right]$. More concisely,

$$\frac{dQ}{d\alpha} = \left[\frac{1}{Q} \left\langle f(T, x) \text{sgnu} \left(Ru + \sqrt{Q^2 - R^2} v \right) \right\rangle + \frac{1}{2Q} \langle f(T, x)^2 \rangle \right],$$
$$\frac{dR}{d\alpha} = \langle f(T, x) (\text{sgnu})(u) \rangle,$$

Generalization Error

- The generalization error ε_g is obtained by comparing the student and teacher outputs, averaged over the distribution of example inputs.
- In the two-dimensional plane of u and v , the teacher outputs $+1$ when $u > 0$ and -1 when $u < 0$.
- Let $\rho = R/Q$. The student outputs $+1$ when $\rho u + \sqrt{1 - \rho^2}v > 0$ and -1 when $\rho u + \sqrt{1 - \rho^2}v < 0$.
- Introducing the step function Θ , we write

$$\varepsilon_g = \int Du \int Dv \Theta \left[-u \left(\rho u + \sqrt{1 - \rho^2}v \right) \right].$$



$$\phi = \operatorname{arccot} \frac{\rho}{\sqrt{1 - \rho^2}} = \arccos \rho$$

Geometric Interpretation

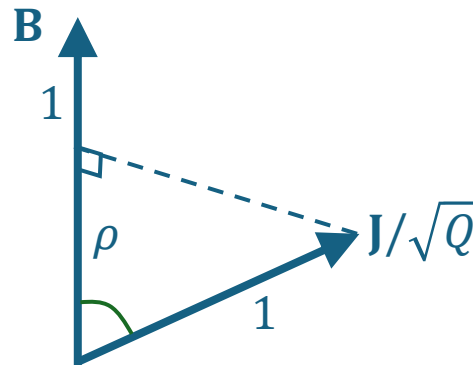
- Transforming to polar coordinates in the two-dimensional plane of u and v ,

$$\int_{-\infty}^{\infty} \frac{du}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} \int_{-\infty}^{\infty} \frac{dv}{\sqrt{2\pi}} e^{-\frac{v^2}{2}} \Theta \left[-u \left(\rho u + \sqrt{1 - \rho^2} v \right) \right] = \underbrace{\int_0^{\infty} dr r e^{-\frac{r^2}{2}}}_1 \left(\int_{\frac{\pi}{2}}^{\frac{\pi}{2}+\phi} \frac{d\theta}{2\pi} + \int_{\frac{3\pi}{2}}^{\frac{3\pi}{2}+\phi} \frac{d\theta}{2\pi} \right)$$

- The generalization error is equal to the fraction of the angular distribution in which the student output disagrees with the teacher output. Therefore,

$$\varepsilon_g = \frac{2\phi}{2\pi} = \frac{1}{\pi} \arccos(\rho).$$

- Geometric interpretation: The generalization error is given by **the angle between the student vector and the teacher vector**.



Hebb Rule

- For Hebb rule, $f(T, x) = 1$. The differential equation for R becomes very simple:

$$\frac{dR}{d\alpha} = \int Du \int Dv (\text{sgn} u) u = \int Du |u| = \frac{2}{\sqrt{2\pi}}.$$

- Solution:

$$R = \frac{2\alpha}{\sqrt{2\pi}}.$$

- This means that the projection of the student vector onto the teacher vector **grows linearly with time**.

Dynamical Equation for Q

$$\frac{dQ}{d\alpha} = \int Du \int Dv \left[\frac{1}{Q} \operatorname{sgnu} \left(Ru + \sqrt{Q^2 - R^2} v \right) + \frac{1}{2Q} \right].$$

- Transforming to polar coordinates,

$$\frac{dQ}{d\alpha} = \int_0^\infty dr r^2 e^{-\frac{r^2}{2}} \left(\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{d\theta}{2\pi} - \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{d\theta}{2\pi} \right) (\rho \cos \theta + \sqrt{1 - \rho^2} \sin \theta) + \frac{1}{2Q} \int_0^\infty dr r e^{-\frac{r^2}{2}}$$

The integrals over r and θ in the first term are decoupled:

$$\begin{aligned} \int_0^\infty dr r^2 e^{-\frac{r^2}{2}} &= \frac{\sqrt{2\pi}}{2}, \\ \left(\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{d\theta}{2\pi} - \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{d\theta}{2\pi} \right) (\rho \cos \theta + \sqrt{1 - \rho^2} \sin \theta) &= \left(\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{d\theta}{2\pi} - \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{d\theta}{2\pi} \right) \cos(\theta - \phi) \\ &= \frac{4 \cos \phi}{2\pi} = \frac{4\rho}{2\pi}. \end{aligned}$$

Asymptotic Generalization Error

- Substituting into the differential equation

$$\frac{dQ}{d\alpha} = \frac{2\rho}{\sqrt{2\pi}} + \frac{1}{2Q} = \frac{4\alpha}{2\pi Q} + \frac{1}{2Q}.$$

- The solution is

$$\frac{Q^2}{2} = \frac{2\alpha^2}{2\pi} + \frac{\alpha}{2}.$$

- In the limit $\alpha \rightarrow \infty$,

$$\begin{aligned} Q &\approx \frac{2\alpha}{\sqrt{2\pi}} \left(1 + \frac{\pi}{4\alpha}\right), \\ \rho &= \frac{R}{Q} \approx 1 - \frac{\pi}{4\alpha}, \\ \varepsilon &= \frac{1}{\pi} \arccos \rho \approx \frac{1}{\pi} \sqrt{2(1 - \rho)} \approx \frac{1}{\sqrt{2\pi}} \alpha^{-\frac{1}{2}}. \end{aligned}$$

Comparing with Off-line Learning etc

- The generalization error vanishes for sufficiently large number of on-line examples.
- However, the generalization error of on-line learning does not converge as fast as off-line learning (batch learning), which is asymptotically [1]

$$\varepsilon_g(\alpha \rightarrow \infty) \approx 0.44\alpha^{-1}.$$

- Other learning rules:
 - Perceptron algorithm: $f(T, x) = \Theta(-Tx): \varepsilon_g(\alpha \rightarrow \infty) \sim \alpha^{-\frac{1}{3}}$
 - Perceptron algorithm with nonzero stability: $f(T, x) = \Theta\left(\kappa - \frac{Tx}{Q}\right): \varepsilon_g(\alpha \rightarrow \infty) \sim \alpha^{-\frac{1}{2}}$ (see next slide)
 - AdaTron algorithms: $f(T, x) = (-Tx)\Theta(-Tx): \varepsilon_g = \frac{3}{2}\alpha^{-1}$ (see next slide)
 - Results may be improved by α -dependent learning rate.

[1] For example, T. L. H. Watkin, Europhys. Lett. **21**, 871 (1993).

Results

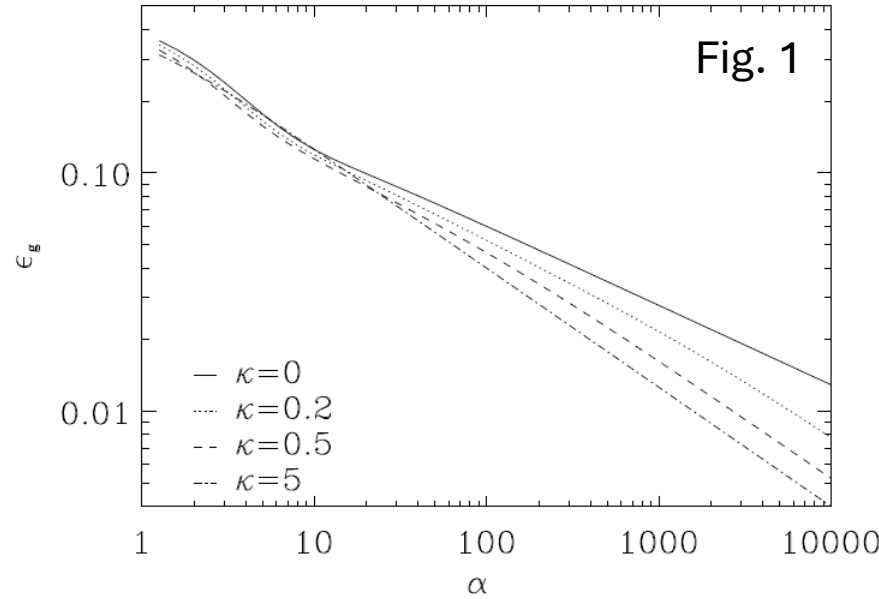


Fig. 1. – Numerical solution of the generalization error for the perceptron learning rule. Curves are shown for different stabilities κ (— $\kappa = 0$, $\kappa = 0.2$, --- $\kappa = 0.5$, -.-.- $\kappa = 5$) and initial conditions $R(0) = 0$, $Q(0) = 1$. The generalization error decays differently on an α -scale for $\kappa = 0$ and $\kappa > 0$. The results for $\kappa = 5$ and Hebbian learning are virtually identical.

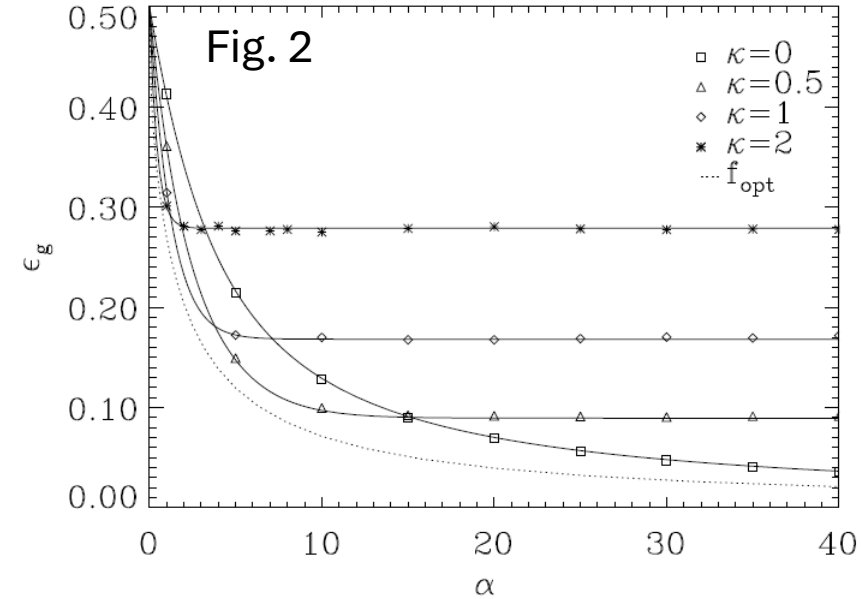


Fig. 2. – Evolution of the generalization error for the AdaTron learning rule for the same initial conditions as in fig. 1. The symbols show results obtained by simulations performed for a perceptron of 100 input neurons averaged over 100 independent runs ($\square$ $\kappa = 0$, $\triangle$ $\kappa = 0.5$, $\diamond$ $\kappa = 1$, $*$ $\kappa = 2$). Standard error bars would be smaller than the size of the symbols. The solid lines represent the numerical solutions of the corresponding differential equation. The dashed line is the numerical solution of the optimal learning process with weight function f_{opt} [9]. Note that the AdaTron learning rule for zero stability performs comparably well.

4. On-line learning in soft committee machines

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The problem of on-line learning in two-layer neural networks is studied within the framework of statistical mechanics. A fully connected committee machine with K hidden units is trained by gradient descent to perform a task defined by a teacher committee machine with M hidden units acting on randomly drawn inputs. The approach, based on a direct averaging over the activation of the hidden units, results in a set of first-order differential equations that describes the dynamical evolution of the overlaps among the various hidden units and allows for a computation of the generalization error. The equations of motion are obtained analytically for general K and M and provide a powerful tool used here to study a variety of realizable, overrealizable, and unrealizable learning scenarios and to analyze the role of the learning rate in controlling the evolution and convergence of the learning process.

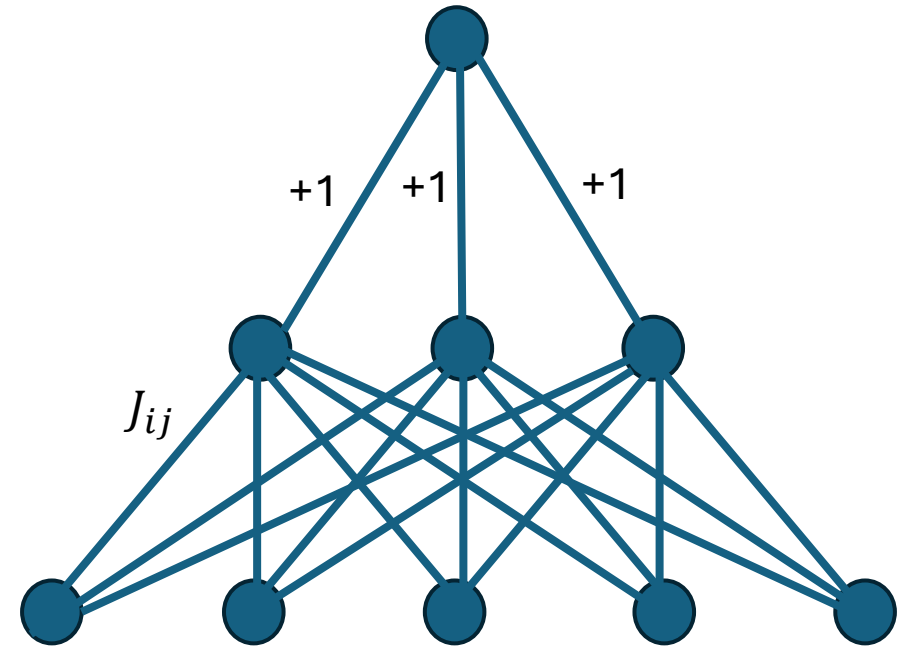
PACS number(s): 87.10.+e, 02.50.-r, 05.20.-y

Soft Committee Machine

- N input units, K hidden units, one linear output unit
- When an input pattern $\xi = (\xi_1, \dots, \xi_N)$ is presented, the field at hidden unit i is $x_i = \mathbf{J}_i \cdot \xi$.
- Hidden-to-output weights are all +1. Overall output

$$S(\mathbf{J}, \xi) = \sum_{i=1}^K g(\mathbf{J}_i \cdot \xi).$$

- g is the activation function ($g(x) = \text{erf}\left(\frac{x}{\sqrt{2}}\right)$ in this paper).



The Teacher Network

- Consider the **teacher network** with weight vector $\mathbf{B}_n = (B_{n1}, \dots, B_{nN})$
- At time step μ , a new, uncorrelated input ξ_j^μ ($j = 1, \dots, N$ and $\mu = 1, \dots, p$) is presented, where ξ_j^μ are taken to be i.i.d. random numbers with zero mean and unit variance.
- The teacher network produces the output

$$T_\mu = \sum_{n=1}^M g(\mathbf{B}_n \cdot \boldsymbol{\xi}^\mu).$$

Teacher and Student Fields Are Gaussian

- Since the components of the example inputs are i.i.d., the fields of the hidden layers of the teacher are Gaussian distributed with covariances

$$\langle y_n y_m \rangle = \left\langle \sum_{j,k} B_{nj} \xi_j^\mu B_{mk} \xi_k^\mu \right\rangle = \sum_j B_{nj} B_{mj} \left\langle (\xi_j^\mu)^2 \right\rangle = \sum_j B_{nj} B_{mj} = T_{nm}.$$

- Similarly, the student fields are Gaussian distributed with covariances

$$\langle x_i x_j \rangle = \left\langle \sum_{k,l} J_{ik} \xi_k^\mu J_{jl} \xi_l^\mu \right\rangle = \sum_k J_{ik} J_{jk} \left\langle (\xi_k^\mu)^2 \right\rangle = \sum_k J_{ik} J_{jk} = Q_{ij}.$$

- The covariance between the student and teacher fields is given by

$$\langle x_i y_n \rangle = \left\langle \sum_{j,k} J_{ij} \xi_j^\mu B_{nk} \xi_k^\mu \right\rangle = \sum_j J_{ij} B_{nj} \left\langle (\xi_j^\mu)^2 \right\rangle = \sum_j J_{ij} B_{nj} = R_{in}.$$

Teacher and Student Field Distribution

- The covariance matrix of student and teacher fields is given by

$$C = \begin{pmatrix} \langle x_i x_j \rangle & \langle x_i y_n \rangle \\ \langle y_n x_i \rangle & \langle y_n y_m \rangle \end{pmatrix} = \begin{pmatrix} Q & R \\ R & T \end{pmatrix}.$$

- The joint distribution of $\mathbf{x}$ and $\mathbf{y}$ is given by

$$P(\mathbf{x}, \mathbf{y}) = \frac{1}{\sqrt{(2\pi)^{M+K} \det C}} \exp \left[-\frac{1}{2} (\mathbf{x}, \mathbf{y})^T C^{-1} (\mathbf{x}, \mathbf{y}) \right].$$

Generalization Error

- Error function:

$$\epsilon(\mathbf{J}, \xi) \equiv \frac{1}{2} [S(\mathbf{J}, \xi) - T(\mathbf{B}, \xi)]^2 = \frac{1}{2} \left[\sum_i^K g(x_i) - \sum_n^M g(y_n) \right]^2 .$$

- Recall that the generalization error of the perceptron, when averaged over i.i.d. inputs, becomes dependent only on the **variance Q** of the student vector and the **covariance R** between the student and teacher vector.

Average Generalization Error

- In the soft committee machine, we have a similar result, namely, the generalization error only depends on the **variances and covariances of the student and teacher vector**:

$$\begin{aligned}\epsilon_g(\mathbf{J}) = & \frac{1}{\pi} \left\{ \sum_{i,k} \arcsin \frac{Q_{ik}}{\sqrt{1+Q_{ii}}\sqrt{1+Q_{kk}}} \right. \\ & + \sum_{n,m} \arcsin \frac{T_{nm}}{\sqrt{1+T_{nn}}\sqrt{1+T_{mm}}} \\ & \left. - 2 \sum_{i,n} \arcsin \frac{R_{in}}{\sqrt{1+Q_{ii}}\sqrt{1+T_{nn}}} \right\}.\end{aligned}$$

Gradient Descent Rule

$$\mathbf{J}^{\mu+1} = \mathbf{J}^{\mu} - \frac{\eta}{N} \nabla_{\mathbf{J}} \epsilon(\mathbf{J}^{\mu}, \boldsymbol{\xi}^{\mu}),$$

Note that the learning rate η scales with the input size N ,

$$\mathbf{J}_i^{\mu+1} = \mathbf{J}_i^{\mu} + \frac{\eta}{N} \delta_i^{\mu} \boldsymbol{\xi}^{\mu},$$

with

$$\delta_i^{\mu} \equiv g'(x_i^{\mu}) \left[\sum_{n=1}^M g(y_n^{\mu}) - \sum_{j=1}^K g(x_j^{\mu}) \right].$$

Dynamical Equation of R^μ

- Change in a single step,

$$\Delta \mathbf{J}_i^\mu = \frac{\eta}{N} \delta_i^\mu \xi^\mu.$$

- Dot product with the teacher vector $\mathbf{B}_n$,

$$\Delta \mathbf{J}_i^\mu \cdot \mathbf{B}_n = \frac{\eta}{N} \delta_i^\mu \xi^\mu \cdot \mathbf{B}_n,$$

$$\Delta R_{in}^\mu = \frac{\eta}{N} \delta_i^\mu y_n^\mu.$$

- In the limit $N \rightarrow \infty$, we introduce the continuous time $\alpha = \frac{\mu}{N}$.

$$\Delta R_{in}^\mu = \Delta \alpha \eta \delta_i^\mu y_n^\mu.$$

$$\frac{dR_{in}}{d\alpha} = \eta \langle \delta_i y_n \rangle.$$

The averages on the right-hand side can be expressed in terms of the variances and covariances of the student and teacher vectors (Q_{ij}, R_{in}, T_{nm}).

Dynamical Equation of Q^μ

- In a single step,

$$\mathbf{J}_i^{\mu+1} = \mathbf{J}_i^\mu + \frac{\eta}{N} \delta_i^\mu \xi^\mu.$$

- Dot product with $\mathbf{J}_k^{\mu+1} = \mathbf{J}_k^\mu + \frac{\eta}{N} \delta_k^\mu \xi^\mu$,

$$\mathbf{J}_i^{\mu+1} \cdot \mathbf{J}_k^{\mu+1} = \mathbf{J}_i^\mu \cdot \mathbf{J}_k^\mu + \frac{\eta}{N} \delta_i^\mu \xi^\mu \cdot \mathbf{J}_k^\mu + \frac{\eta}{N} \delta_k^\mu \xi^\mu \cdot \mathbf{J}_i^\mu + \frac{\eta^2}{N^2} \delta_i^\mu \delta_k^\mu \xi^\mu \cdot \xi^\mu,$$

$$Q_{ik}^{\mu+1} = Q_{ik}^\mu + \frac{\eta}{N} (\delta_i^\mu x_k^\mu + \delta_k^\mu x_i^\mu) + \frac{\eta^2}{N} \delta_i^\mu \delta_k^\mu,$$

$$\Delta Q_{ik} = \frac{1}{N} [\eta(\delta_i x_k + \delta_k x_i) + \eta^2 \delta_i \delta_k].$$

- In the limit $N \rightarrow \infty$, we introduce the continuous time $\alpha = \frac{\mu}{N}$.

$$\frac{dQ_\mu}{d\alpha} = \eta \langle \delta_i x_k \rangle + \eta \langle \delta_k x_i \rangle + \eta^2 \langle \delta_i \delta_k \rangle.$$

- The averages on the right-hand side can be expressed in terms of the variances and covariances of the student and teacher vectors (Q_{ij}, R_{in}, T_{nm}) .⁵²

Results

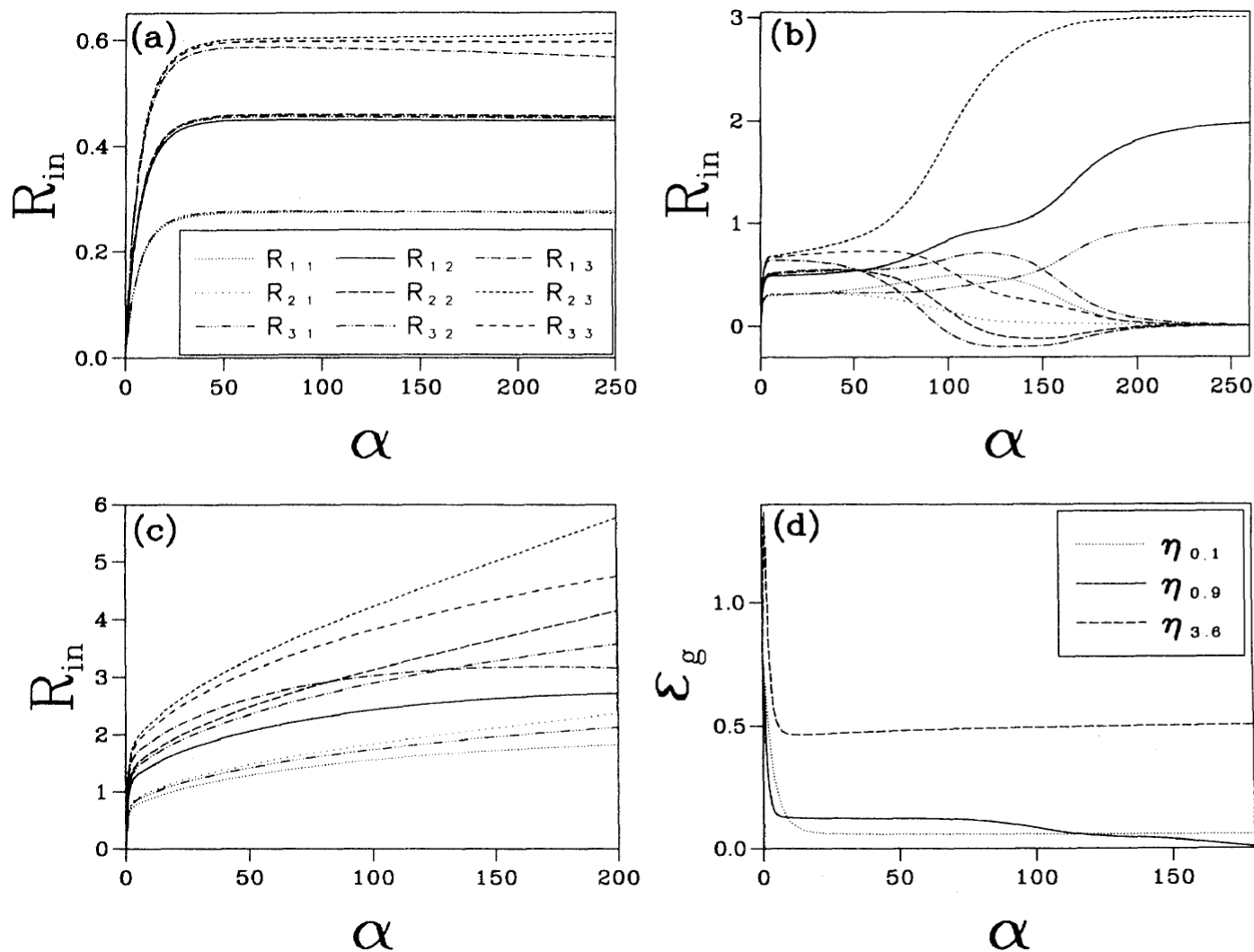


FIG. 1. Dependence of the overlaps between various student and teacher vectors and the generalization error on the normalized number of examples α , for several values of the learning rate η . The overlaps are shown for (a) $\eta = 0.1$, (b) $\eta = 0.9$, and (c) $\eta = 3.6$. The generalization error for these three cases is shown in (d). The teacher is characterized by $T_{nm} = n\delta_{nm}$. Initial conditions are $R_{in} = 0$ and $Q_{ik} = U[0, 0.5]\delta_{ik}$.

The Symmetric Plateau

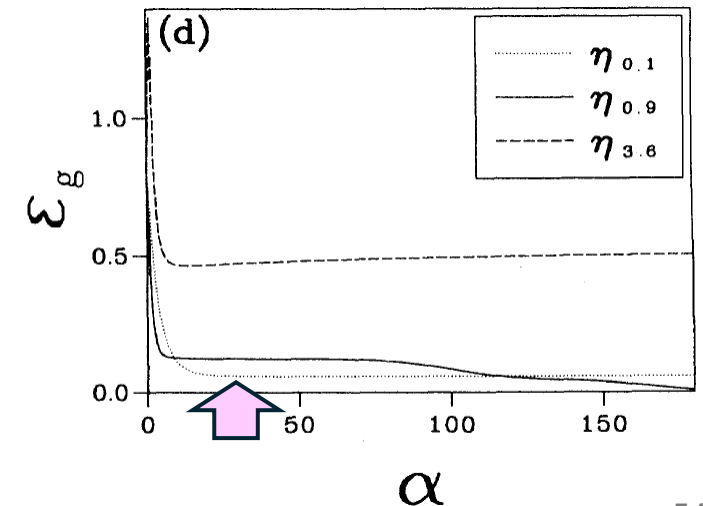
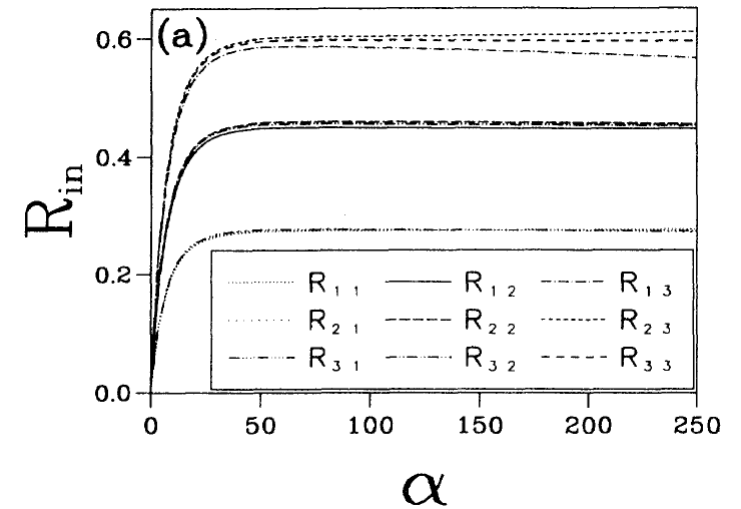
- Low learning rate ($\eta = 0.1$ in Fig. 1(a)).

$$R_{11} \approx R_{21} \approx R_{31},$$

$$R_{12} \approx R_{22} \approx R_{32},$$

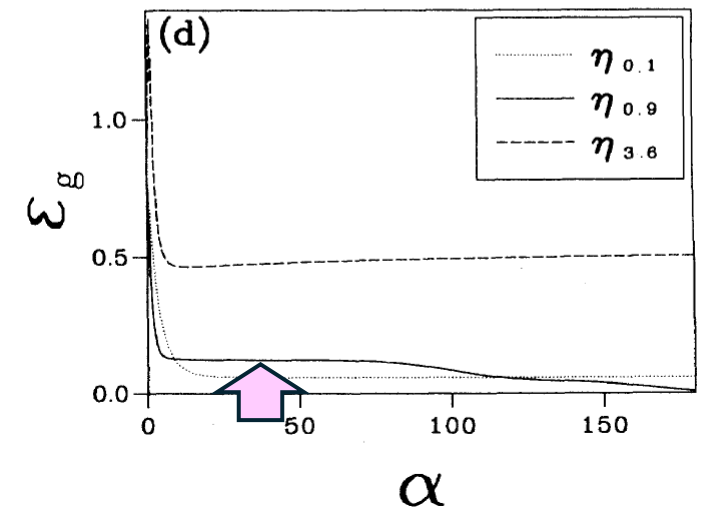
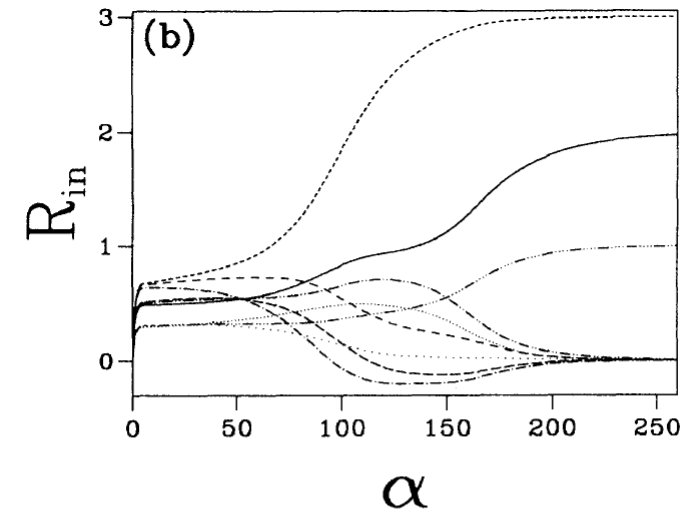
$$R_{13} \approx R_{23} \approx R_{33}.$$

- The three student vectors have the **same overlaps** with the three teacher vectors.
- The student vectors are **identical** to each other.
- The student **cannot differentiate** the teacher hidden units.
- Hence, the student network is in the **symmetric** regime.
- The generalization error remains **finite**.



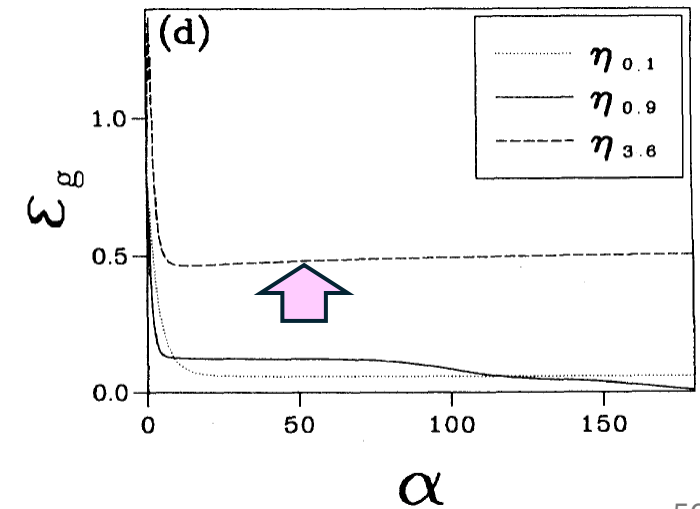
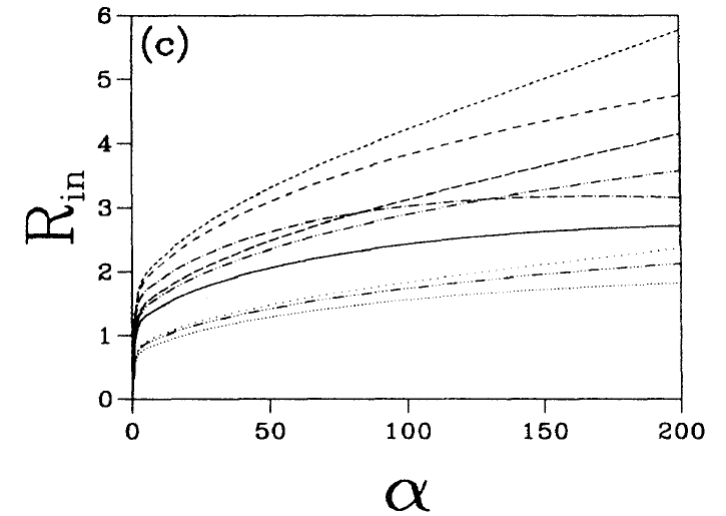
The Specialization Regime

- Intermediate learning rate ($\eta = 0.9$ in Fig. 1(b)).
- In the early stage, the overlaps R_{in} lie in the symmetric subspace.
- Afterwards, the student units become **specialized** (each student vector has an especially large overlap with a teacher vector).
- Eventually, the matrix $\mathbf{R}$ becomes identical to the matrix $\mathbf{T}$.
- Final states obtained by **permuting** the labels of the hidden units are possible.
- The generalization error approaches zero.



The Divergent Regime

- High learning rate ($\eta = 3.6$ in Fig. 1(c)).
- Uncontrolled growth of the student vectors.
- No specialization.
- The generalization error is nonzero.



Example Size Requirement

- Here, α^* is the number of examples required to reach $\epsilon_g = 0.01$.
- α^* diverges at small η due to trapping in the **symmetric** phase.
- When η increases, α^* decreases $\Rightarrow$ **faster** convergence to the optimal solution.
- At the **optimal learning rate** η_{opt} , α^* is minimum.
- When $\eta > \eta_{\text{opt}}$, α^* increases and **diverges** as $\eta \rightarrow \eta_{\text{max}}$.
- Both η_{opt} and η_{max} decreases with the number K of **hidden units**.
- $\alpha_{\text{opt}} = \alpha^*(\eta_{\text{opt}})$ increases rapidly with K .

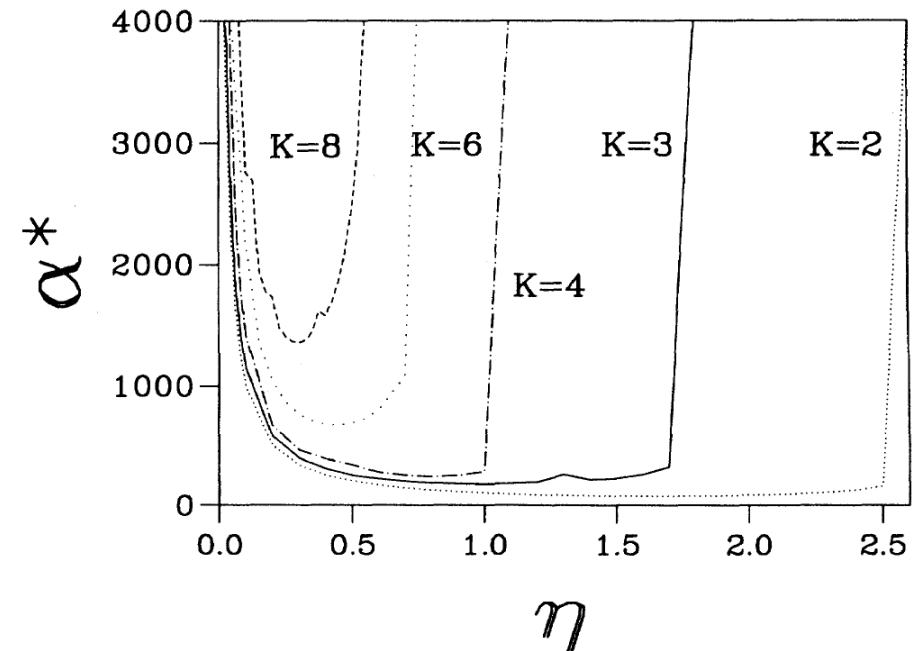


FIG. 2. Dependence of the number of examples α^* needed to achieve $\epsilon_g = 0.01$ on the learning rate η for several values of the number $K = M$ of hidden units.

Summary of Concepts

- History
 - Synapses and weights
 - Learning
- Learning dynamics
 - Gradient descent
 - Boltzmann distribution
 - Temperature effects
- Online learning
 - Student vector magnitude and student-teacher overlap
 - Generalization error
- Soft committee
 - Symmetric regime
 - Specialization regime
 - Divergent regime